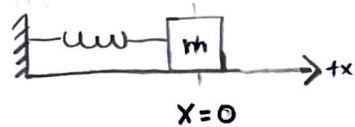


Small Oscillations: Simple Harmonic Oscillator

We will consider the oscillatory motion of a particle in 1D.



$x=0$: Equilibrium point

restoring force

$F(x) = -kx$: Hooke's law : approximation!

displacement

We assume that the displacements of the particle (block) are sufficiently small such that we can neglect all the other terms in the F function with a higher power than x .

$$F = m\ddot{x}$$

$$-kx = m\ddot{x} \rightarrow \ddot{x} + \left(\frac{k}{m}\right)x = 0 \equiv \omega_0^2$$

$$\Rightarrow \ddot{x} + \omega_0^2 x = 0$$

We know that $x(t) = e^{\lambda t}$ is a solution for this equation.

$$\dot{x}(t) = \lambda e^{\lambda t}$$

$$\ddot{x}(t) = \lambda^2 e^{\lambda t}$$

insert $\ddot{x}(t)$ and $x(t)$ into the eq. of motion.

$$\lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0$$

$$\lambda^2 = -\omega_0^2 \Rightarrow \lambda_{1,2} = \pm i\omega_0$$

$$\Rightarrow x_1(t) = e^{+i\omega_0 t}$$

$$x_2(t) = e^{-i\omega_0 t}$$

general solution $\rightarrow x(t) = A e^{+i\omega_0 t} + B e^{-i\omega_0 t}$

There is a problem with these initial conditions

$$x(t) = A e^{+i\omega_0 t} + B e^{-i\omega_0 t}$$

initial conditions.

(C, D ∈ ℝ)

When we insert the initial conditions, which are real numbers, into the solution, it turns out to be a real solution!

$$x(0) = A + B = C$$

$$\dot{x}(0) = A i\omega_0 e^{i\omega_0 \cdot 0} - B i\omega_0 e^{-i\omega_0 \cdot 0} = D$$

$$\Rightarrow i\omega_0(A - B) = D$$

$$D = i\omega_0 A - i\omega_0 B \Rightarrow \frac{D}{i\omega_0} = A - B$$

$$C = A + B$$

$$A = \frac{1}{2} \left(C + \frac{D}{i\omega_0} \right)$$

$$B = \frac{1}{2} \left(C - \frac{D}{i\omega_0} \right)$$

$$x(t) = \frac{1}{2} \left(C + \frac{D}{i\omega_0} \right) e^{i\omega_0 t} + \frac{1}{2} \left(C - \frac{D}{i\omega_0} \right) e^{-i\omega_0 t}$$

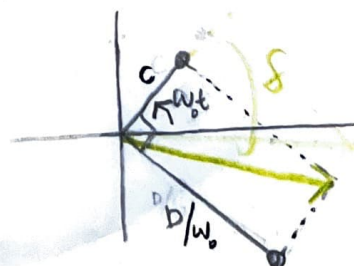
$$(\cos \omega_0 t + i \sin \omega_0 t) \quad (\cos \omega_0 t - i \sin \omega_0 t)$$

$$= \frac{1}{2} \left\{ C \cos \omega_0 t + i C \sin \omega_0 t + \frac{D}{i\omega_0} \cos \omega_0 t + \frac{D}{i\omega_0} i \sin \omega_0 t \right.$$

$$\left. C \cos \omega_0 t - i C \sin \omega_0 t - \frac{D}{i\omega_0} \cos \omega_0 t + \frac{D}{i\omega_0} i \sin \omega_0 t \right\}$$

$$= C \cos \omega_0 t + \frac{D}{\omega_0} \sin \omega_0 t = C \cos \omega_0 t + \frac{D}{\omega_0} \cos \left(\omega_0 t - \frac{\pi}{2} \right)$$

$$= F \cos(\omega_0 t - \delta)$$



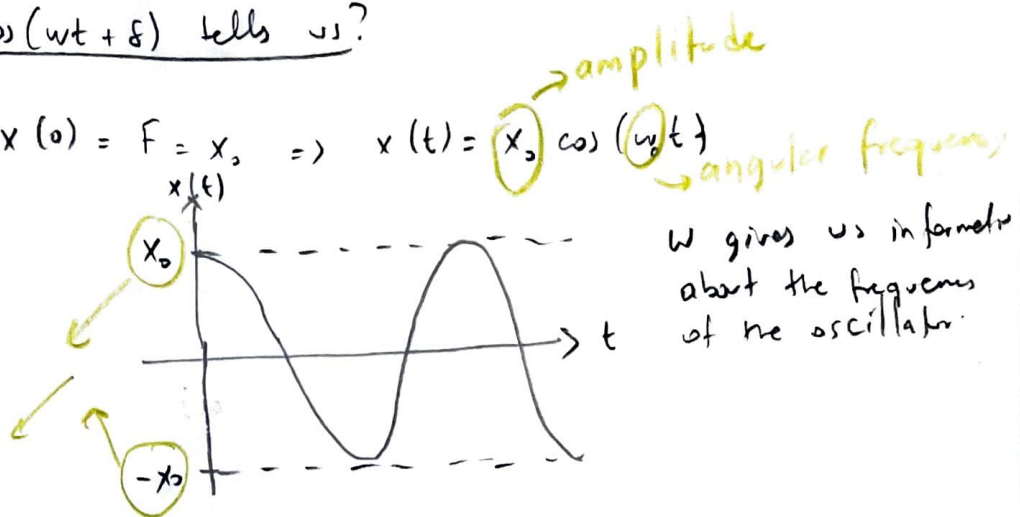
This solution can also be reprinted in another form: $x(t) = F \sin(\omega_0 t + \phi)$

What does $x(t) = F \cos(\omega t + \delta)$ tell us?

$$x(0) = X_0 \Rightarrow x(0) = F = X_0 \Rightarrow x(t) = X_0 \cos(\omega t)$$

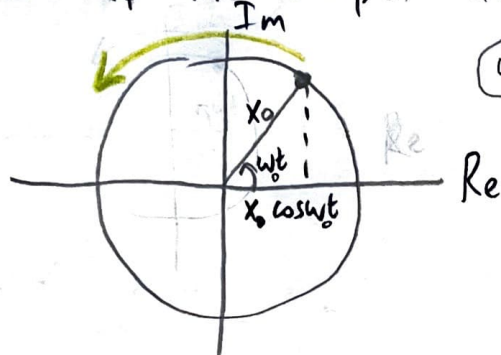
$\delta = 0$: It is all about when we started our time.

The max distance that a SHO can reach from its equilibrium point.



Describing the dynamics on the complex plane:
(or relationship b/w harmonic osc. and circular motion).

Think of the complex number $z(t) = X_0 e^{i\omega t}$



ω : How much angle is swept in the complex plane in unit time?

$$= X_0 [\cos \omega t + i \sin \omega t]$$

$$= X_0 \cos \omega t + i X_0 \sin \omega t$$

$$x(t) = \text{Re}[z(t)]$$

δ only determines the "phase" of the motion

★ The benefit of complex representation is that linear operations with other complex representations produces a complex result whose real part reflects the same linear operations with the real parts of other complex sinusoids.

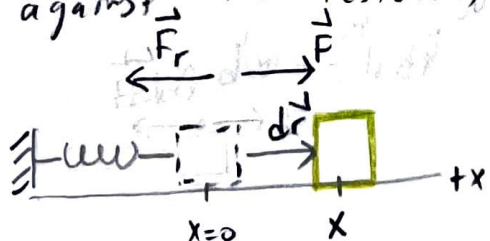
Energy of the simple harmonic oscillator

$$T(t) = \frac{1}{2} m \dot{x}^2 = \frac{1}{2} m X_0^2 \omega^2 \sin^2(\omega t - \delta) \quad ; \text{Kinetic energy.}$$

$$x(t) = X_0 \cos(\omega t - \delta)$$

$$\dot{x}(t) = -X_0 \omega \sin(\omega t - \delta)$$

• The potential energy may be obtained by calculating the work required to displace the particle a distance x . The incremental amount of work dW necessary to move the particle by an amount dx against the restoring force F_r is



We are interested in the work done by us. Therefore $\vec{F} \cdot d\vec{r}$ is positive; the work done by us must be positive:

$$dW = + \vec{F}_r \cdot d\vec{r} = + kx dx$$

(Kittel)

$$dw = kx dx \rightarrow W = \int_0^x kx dx = \frac{1}{2} k x^2 \equiv U : \text{potential energy of the simple harmonic oscillator.}$$

The time dependence of potential energy:

$$U(t) = \frac{1}{2} k x_0 \cos^2(\omega_0 t - \delta)$$

Therefore the total energy of SHO at any time t can be expressed as

$$E(t) = T(t) + U(t) = \frac{1}{2} m x_0^2 \omega_0^2 \sin^2(\omega_0 t - \delta) + \frac{1}{2} k x_0 \cos^2(\omega_0 t - \delta)$$

Remember that $\omega_0^2 = k/m$

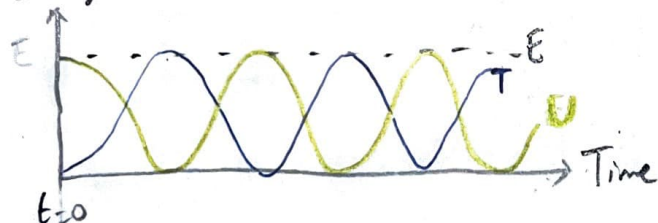
$$\Rightarrow E(t) = \frac{1}{2} x_0^2 \cancel{\omega_0^2} \frac{k}{\cancel{m}} \sin^2(\omega_0 t - \delta) + \frac{1}{2} k x_0 \cos^2(\omega_0 t - \delta)$$

$$= \frac{1}{2} x_0^2 k \left\{ \sin^2(\omega_0 t - \delta) + \cos^2(\omega_0 t - \delta) \right\} = \frac{1}{2} k x_0^2$$

★ The total energy of SHO is proportional to the square of the amplitude.

★ E is independent of the time; energy is conserved.

Energy



! It is independent of the amplitude (or total energy).

Period, frequency, angular frequency

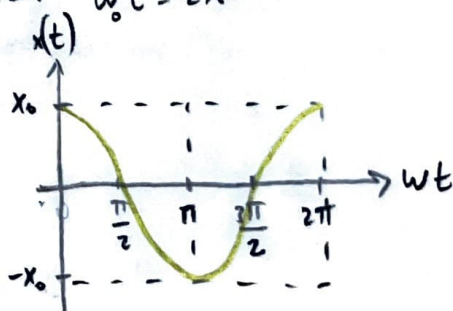
Period (τ_0): Time interval between the successive repetitions of the particle's position. Unit: [T] SI: s (seconds)

$x(t) = x_0 \cos(\omega_0 t - \delta)$ repeats itself when $\omega_0 t = 2\pi$

$$\Rightarrow \omega_0 \tau_0 = 2\pi \Rightarrow \tau_0 = 2\pi / \omega_0 = 2\pi \sqrt{\frac{m}{k}}$$

Frequency (ν_0): How many times does the oscillators motion repeats itself in unit time?

frequency: [vibrations per unit time] $\rightarrow \omega_0 \equiv 2\pi \nu_0 \Rightarrow \nu_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$
angular frequency: [radians per unit time]



ω_0 $1/[T]$: SI $\frac{1}{s} \text{ rad}$

ν_0 $1/[T]$: SI $1/s = 1 \text{ Hz}$

Why almost "everything" is a simple harmonic oscillator?

We saw that the potential of SHO is $U(x) = \frac{1}{2} kx^2$, which involves only the second power of displacement. Also, the restoring force $F(x)$ only involves the first power of displacement x , as also can be seen by the relation

$$F = - \frac{dU(x)}{dx} = -kx$$

However, in general the potential of a given random physical system can be any continuous function of x . Therefore, since we know that any continuous function can be written in terms of a Taylor expansion:

$$U(x) = U(x_0) + U'(x_0)(x-x_0) + \frac{U''(x_0)}{2!}(x-x_0)^2 + \frac{U'''(x_0)}{3!}(x-x_0)^3 + \dots$$

Let us assume that x_0 is the equilibrium point for the particle. Then, we know that $U'(x_0) = 0$. Let's also assume that x_0 is also stable; $U''(x_0) > 0$. So,

$$U(x) = U(x_0) + \frac{U''(x_0)}{2!}(x-x_0)^2 + \frac{U'''(x_0)}{3!}(x-x_0)^3 + \dots$$

We can see from this expression that choosing the restoring force as $F = -kx$ is just an assumption. In general, the force can be more complicated in terms of containing higher powers of x for the systems going through an oscillatory motion since $F = -\frac{d}{dx}[U(x)]$. However, if we assume that $x \ll 1$, then we can write the potential in the form,

$$U(x) \approx U(x_0) + \frac{U''(x_0)}{2}(x-x_0)^2$$

Keeping in mind that we can choose $x_0 \equiv 0$ and $U(x_0) \equiv 0$, the above equation gives us nothing but the same potential of SHO:

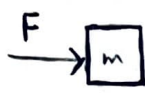
$$U(x) = \frac{1}{2} U''(0) x^2 \quad \text{where in SHO } U''(0) = k$$

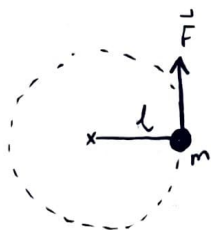
Therefore, it does not matter what kind of $U(x)$ we have (or equivalently what kind of physical system we are considering), all the systems oscillating around a stable fixed point look like a SHO if we assume that $x \ll 1$.

The oscillation (angular) frequency is given by

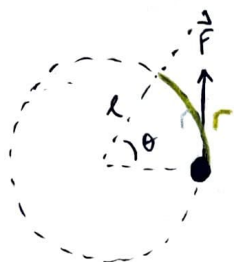
$$\omega = \sqrt{\frac{U''(x_0)}{m}}$$

Newton's second law in the rotational form

 : A force applied on a particle simply accelerates it in 1D.
 $F = ma$



If a tangential force is applied to a rotating body, it also accelerates it, but this time acceleration takes place on a circle. So, the tangential force changes the particle's angular velocity in time. One may intuitively expect that there must be a relationship between time rate of change of ω and the torque N : $N \leftrightarrow \dot{\omega}$



$$F = m \frac{d^2 r}{dt^2}$$

By the definition of 1 rad, we can write the below identity:

$$\left. \begin{array}{l} 1 \text{ rad} \rightarrow l \\ \theta \rightarrow r \end{array} \right\} \theta = \frac{r}{l} (\text{rad}) \Rightarrow r = \theta l$$

$$F = m \frac{d^2(\theta l)}{dt^2} = ml \frac{d^2 \theta}{dt^2} ; \text{ we already defined } \omega \equiv \frac{d\theta}{dt} \text{ angular velocity}$$

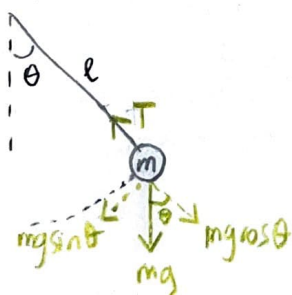
★ Therefore, the time rate of change of the angular velocity is called angular acceleration, denoted by α : $\alpha \equiv \dot{\omega}$

$\Rightarrow F = ml\alpha \rightarrow \textcircled{lF} = \textcircled{ml^2\alpha} \Rightarrow \boxed{N = I\alpha}$ rotational form of Newton's second law.

N : torque
 I : rotational inertia.

★ Therefore, rotational inertia behaves like mass and torque behaves like force in Newton's second law, when we consider the angular acceleration of a rotating body. (mathematical)

Yet another example of SHM: Simple pendulum (will be considered as an example)



We can divide the gravitational force acting on the body into 2 components. One component is in the direction of T , the tension on the string. It does not cause any motion:

$$T = mg \cos \theta$$

The other component makes the pendulum oscillate around its equilibrium point $\theta = 0$. This component of the gravitational force is tangential to pendulum's motion. we will make use of Newton's 2nd law in its rotational form to study pendulum problem because the component $mg \sin \theta$ applies a torque on the mass m and changes its angular velocity in time.

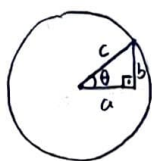
$$N = I\alpha$$

$$lmg \sin \theta = -ml^2 \ddot{\theta}$$

In above equation all l, m, g are positive by definition. As the pendulum moves left θ and $\sin \theta$ decreases but their values are positive. The angular velocity ω increases in magnitude, however its value is negative since θ decreases, so $\dot{\omega}$ is.

Therefore, to make $lmg \sin \theta$ and $ml^2 \ddot{\omega}$ equivalent we need to multiply one side with -1 . We will assume that the pendulum makes very small oscillations, $\theta \ll 1$, as we assumed for the harmonic oscillator that $x \ll 1$.

$\sin \theta \approx \theta$ since
(heuristic approach)



$\sin \theta = b/c$: as θ gets smaller, b gets closer to the arc length seen by the central angle θ , so that the equation $\sin \theta = b/c$ starts to behave like $\theta = b/c$

Taylor series approach: $\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$ for small θ , we can neglect the terms higher than θ .

$\Rightarrow \sin \theta \approx \theta$ for $\theta \ll 1$.

$\Rightarrow \cancel{lm} g \sin \theta = -\cancel{ml}^2 \ddot{\omega} \Rightarrow g \theta \approx -l \ddot{\omega} \rightarrow \boxed{\frac{d^2 \theta}{dt^2} \approx -\frac{g}{l} \theta}$ Equation of motion for the pendulum.

Analogy with SHO: $\frac{d^2 x}{dt^2} = -\frac{k}{m} x$: This is exactly the same equation with pendulum except the constants.

Therefore: $\omega_0 = \sqrt{\frac{g}{l}}$: angular velocity of the oscillation.

$\tau = 2\pi \sqrt{\frac{l}{g}}$: period of the oscillation.

and $\theta(t) = \theta_0 \cos(\omega_0 t - \phi)$

$\dot{\theta}(t) = -\theta_0 \omega_0 \sin(\omega_0 t - \phi)$

$\ddot{\theta}(t) = -\theta_0 \omega_0^2 \cos(\omega_0 t - \phi)$

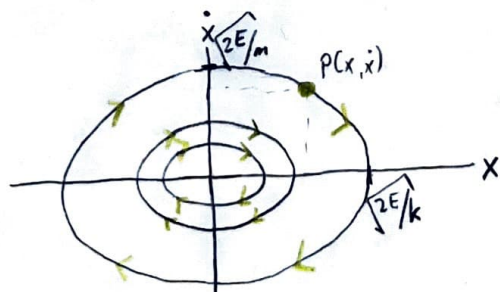
Phase diagrams of the oscillative behavior:

The state of motion of a harmonic oscillator is completely specified as a function of time if two quantities are given at one instance of time: the initial conditions $x(t_0)$ and $\dot{x}(t_0)$. We may consider these quantities to be coordinates in 2D space, called phase space, as follows:

$$x(t) = x_0 \cos(\omega_0 t - \delta) \rightarrow x^2(t) = x_0^2 \cos^2(\omega_0 t - \delta)$$

$$\dot{x}(t) = -x_0 \omega_0 \sin(\omega_0 t - \delta) \rightarrow \dot{x}^2(t) = x_0^2 \omega_0^2 \sin^2(\omega_0 t - \delta)$$

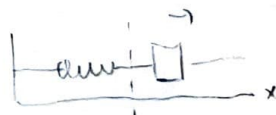
$$\frac{x^2}{x_0^2} = \cos^2(\omega_0 t - \delta) \quad \text{and} \quad \frac{\dot{x}^2}{x_0^2 \omega_0^2} = \sin^2(\omega_0 t - \delta) \Rightarrow \boxed{\frac{x^2}{x_0^2} + \frac{\dot{x}^2}{x_0^2 \omega_0^2} = 1}$$
 Equation of family of ellipses



$$E = \frac{1}{2} k x_0^2 \quad \omega_0^2 = k/m$$

$$\Downarrow$$

$$\frac{x^2}{2E/k} + \frac{\dot{x}^2}{2E/m} = 1$$



Properties of phase diagrams

- i) Each path corresponds to a definite total energy of the oscillator.
- ii) Two paths never cross. If this would happen, it would mean that a given set of initial conditions $x(t_0)$ and $\dot{x}(t_0)$ belong to two paths. But this is impossible since the solution of the diff. eq for that pair is unique.
- iii) The motion of the point is always clockwise direction.

Damped oscillations:

If the oscillator is subject to some frictional force the dynamics of the oscillator may change in different ways. The frictional force is usually proportional to the body's velocity and is usually written $-b\dot{x}$ with $b > 0$. Therefore, the equation of motion becomes:

(drag force) $\sim -b\dot{x} \rightarrow$ as if it was moving in a fluid.

$$m\ddot{x} = -kx - b\dot{x} \rightarrow m\ddot{x} + kx + b\dot{x} = 0$$

$$\rightarrow \ddot{x} + \frac{k}{m}x + \frac{b}{m}\dot{x} = 0 \rightarrow \ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0 \text{ where } \beta = \frac{b}{2m}, \omega_0 = \sqrt{k/m}$$

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$$

$$\uparrow x(t) = e^{\lambda t}, \dot{x}(t) = \lambda e^{\lambda t}, \ddot{x}(t) = \lambda^2 e^{\lambda t}$$

$$\lambda^2 e^{\lambda t} + 2\beta\lambda e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0 \rightarrow \lambda^2 + 2\beta\lambda + \omega_0^2 = 0 \rightarrow \lambda_{1,2} = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega_0^2}}{2} = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

general solution of the quadratic equation.

$$\Rightarrow x_1(t) = e^{(-\beta + \sqrt{\beta^2 - \omega_0^2})t} \text{ and } x_2(t) = e^{(-\beta - \sqrt{\beta^2 - \omega_0^2})t}$$

The general solution: $x(t) = A_1 x_1(t) + A_2 x_2(t)$

$$= e^{-\beta t} \left\{ A_1 e^{\sqrt{\beta^2 - \omega_0^2} t} + A_2 e^{-\sqrt{\beta^2 - \omega_0^2} t} \right\}$$

Therefore, the solution has 3 possible forms depending on the value of $\beta^2 - \omega_0^2$:

i) $\omega_0^2 > \beta^2$: Underdamping

ii) $\omega_0^2 = \beta^2$: Critical damping

iii) $\omega_0^2 < \beta^2$: Overdamping

Underdamped motion: $\omega_0^2 > \beta^2$

$$\omega_0^2 - \beta^2 = \omega_1^2 : \text{positive. eg. } \sqrt{\beta^2 - \omega_0^2} = \sqrt{-\omega_1^2} = i\omega_1$$

$$\Rightarrow x(t) = e^{-\beta t} \{ A_1 e^{i\omega_1 t} + A_2 e^{-i\omega_1 t} \} \text{ or } x(t) = A e^{-\beta t} \cos(\omega_1 t - \delta)$$

ω_1 : "angular frequency" (Strictly speaking, we cannot define a frequency for damped osc. since the motion is not periodic. However, ω_1 has meaning for a given time period.)

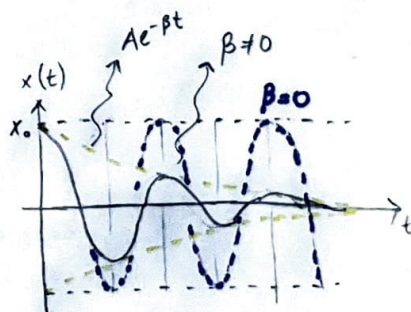
Initial conditions: x_0 (for $\delta=0$)

$$x(0) = x_0 = A$$

$$\Rightarrow x(t) = x_0 e^{-\beta t} \cos(\omega_1 t)$$

★ $\omega_1 < \omega_0$

★ If the damping is small $\omega_1 = \sqrt{\omega_0^2 - \beta^2} \approx \omega_0$



Critically damped motion: $\omega_0^2 = \beta^2$

The roots λ_1 and λ_2 of the auxiliary function are equal. Therefore, two linearly independent solutions, and the general solution becomes:

$$\lambda_{1,2} = -\beta \Rightarrow x_1(t) = e^{-\beta t} \text{ and } x_2(t) = t e^{-\beta t}$$

$$x(t) = A_1 e^{-\beta t} + A_2 t e^{-\beta t} = (A_1 + A_2 t) e^{-\beta t}$$

with initial condition $\rightarrow x(t) = x_0 e^{-\beta t} + \beta x_0 t e^{-\beta t} = x_0 e^{-\beta t} (1 + \beta t)$

Initial conditions: $x(0) = x_0, \dot{x}(0) = 0$

$$x(0) = A_1 = x_0$$

$$\dot{x}(t) = \frac{d}{dt} [A_1 e^{-\beta t} + A_2 t e^{-\beta t}]$$

$$= -\beta A_1 e^{-\beta t} + A_2 (e^{-\beta t} - \beta t e^{-\beta t})$$

$$\dot{x}(0) = -\beta A_1 + A_2 = -\beta x_0 + A_2 = 0$$

$$A_2 = \beta x_0$$

Overdamped motion: $\omega_0^2 < \beta^2$

$$\sqrt{\beta^2 - \omega_0^2} = \omega_2 \rightarrow \text{The roots of the auxiliary function are real.}$$

$$\beta^2 - \omega_0^2 = \omega_2^2$$

$$\lambda_{1,2} = -\beta \pm \sqrt{\beta^2 - \omega_0^2} = -\beta \pm \omega_2$$

$$\Rightarrow x_1(t) = e^{-\beta t} e^{\omega_2 t} \text{ and } x_2(t) = e^{-\beta t} e^{-\omega_2 t} \Rightarrow x(t) = e^{-\beta t} (A_1 e^{\omega_2 t} + A_2 e^{-\omega_2 t})$$

★ ω_2 cannot be regarded as an angular frequency anymore for there is no oscillatory motion at all.

Initial conditions: $x(0) = A_1 + A_2 = x_0$

$$\dot{x}(t) = -\beta e^{-\beta t} (A_1 e^{\omega_2 t} + A_2 e^{-\omega_2 t}) + e^{-\beta t} (A_1 \omega_2 e^{\omega_2 t} - A_2 \omega_2 e^{-\omega_2 t})$$

$$\dot{x}(0) = -\beta (A_1 + A_2) + \omega_2 (A_1 - A_2) = 0$$

$$= -\beta x_0 + \omega_2 (A_1 - A_2) = 0 \Rightarrow \omega_2 (A_1 - A_2) = \beta x_0 \Rightarrow A_1 - A_2 = \frac{\beta x_0}{\omega_2}$$

$$2A_2 = x_0 - \frac{\beta x_0}{\omega_2}$$

$$A_2 = \frac{x_0}{2} - \frac{\beta x_0}{2\omega_2}$$

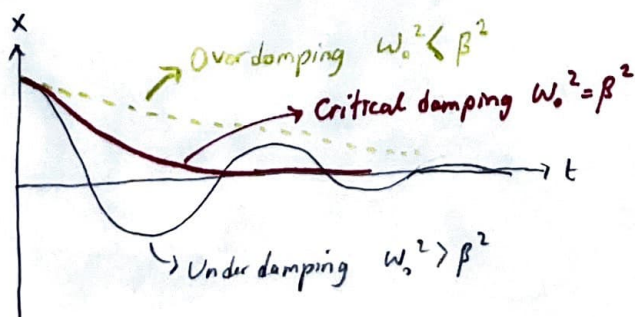
$$= \frac{x_0}{2} \left(1 - \frac{\beta}{\omega_2}\right)$$

$$\Rightarrow 2A_1 = x_0 + \frac{\beta x_0}{\omega_2} \Rightarrow A_1 = \frac{x_0}{2} + \frac{\beta x_0}{2\omega_2} = \frac{x_0}{2} \left(1 + \frac{\beta}{\omega_2}\right)$$

$$\Rightarrow \frac{x_0}{2} + \frac{\beta x_0}{2\omega_2} + A_2 = x_0 \Rightarrow A_2 = \frac{x_0}{2} - \frac{\beta x_0}{2\omega_2} = \frac{x_0}{2} \left(1 - \frac{\beta}{\omega_2}\right)$$

with initial conditions $\rightarrow x(t) = e^{-\beta t} \frac{x_0}{2} \left\{ \left(1 + \frac{\beta}{\omega_2}\right) e^{\omega_2 t} + \left(1 - \frac{\beta}{\omega_2}\right) e^{-\omega_2 t} \right\}$

Comparison of 3 damped behavior:



★ Critical damped motion reaches the equilibrium faster than the others.

e.g. pneumatic-tube screen-door closure system.
car's suspension

→ when a car goes over a bump, the passenger in the car quickly and smoothly regains equilibrium.

★ Emphasize the interplay between

$$\beta = \frac{b}{2m} \quad \omega_0 = \sqrt{\frac{k}{m}}$$

Why critical damping vanishes faster than overdamping?

• $x(t) = x_0 e^{-\beta t} (1 + \beta t)$ $\beta \equiv b/2m$ for critical damping ($\omega_0^2 = \beta^2$)

• $x(t) = x_0 \frac{e^{-\beta t}}{2\omega_z} \left[(\omega_0 + \beta) e^{\omega_z t} + (\omega_0 - \beta) e^{-\omega_z t} \right]$ for overdamping ($\omega_0^2 < \beta^2$)

$$= x_0 e^{-\beta t} \underbrace{\frac{(\omega_0 + \beta)}{2\omega_z} e^{\omega_z t}}_{\text{This term obviously vanishes faster than } x_0 e^{-\beta t} (1 + \beta t)} + x_0 e^{-\beta t} \underbrace{\frac{(\omega_0 - \beta)}{2\omega_z} e^{-\omega_z t}}_{\text{This term obviously vanishes faster than } x_0 e^{-\beta t} (1 + \beta t)}$$

$$e^{\omega_z t} = 1 + \omega_z t + \frac{\omega_z^2 t^2}{2!} + \frac{\omega_z^3 t^3}{3!} + \dots$$

$$\frac{\omega_0 + \beta}{2\omega_z} \left\{ x_0 e^{-\beta t} + x_0 e^{-\beta t} \frac{\omega_z^2 t^2}{2!} + x_0 e^{-\beta t} \frac{\omega_z^3 t^3}{3!} + \dots \right\}$$

↑ compare

$$x_0 e^{-\beta t} + x_0 e^{-\beta t} \beta t$$

In overdamping there are higher order contributions to the result.

Driven oscillations: Sinusoidal Driving forces

We will consider now that the particle which does oscillatory motion is also subjected to an ~~external~~ ^{external} periodic force. This kind of oscillations are called driven oscillations.

e.g. rocking someone on a swing: Swing has its own natural frequency of oscillation, i.e., if someone gives swing an initial push and let it rock on its own, the swing has a natural frequency of oscillation as we have already studied in the pendulum example. Now, we will study what would happen if we keep pushing it with a certain pace, or frequency.

Therefore, the particle is subjected to 3 different forces:

- i) Restoring force $F_r = -kx$
- ii) Damping force $F_d = -b\dot{x}$
- iii) Driving force $F_0 = F_0 \cos \omega t$

This driving force can be another general function $f(x)$. However, for now, we will simply consider sinusoidal driving forces given in above form. The total force acting on the particle, and the equation of motion becomes:

$$F_T = -kx - b\dot{x} + F_0 \cos \omega t \rightarrow \text{Total force}$$

$$m\ddot{x} = -kx - b\dot{x} + F_0 \cos \omega t$$

or in a more familiar form:

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega t \rightarrow \text{Equation of motion.}$$

The same equation can be written as

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = \left(\frac{F_0}{m}\right) \cos \omega t, \text{ where } 2\beta = b/m \text{ and } \omega_0^2 = k/m$$

and ω is the angular frequency of the driving force.

The solution of this nonlinear differential equation has two parts:

i) Complementary solution: It can be obtained by setting RHS zero.

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0, \quad x(t) = e^{\lambda t}, \quad \dot{x}(t) = \lambda e^{\lambda t} \text{ and } \ddot{x}(t) = \lambda^2 e^{\lambda t}$$

$$\Rightarrow \lambda^2 + 2\beta\lambda + \omega_0^2 = 0 \Rightarrow \lambda_{1,2} = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

$$\Rightarrow x_c(t) = e^{-\beta t} \left[A_1 \exp(\sqrt{\beta^2 - \omega_0^2} t) + A_2 \exp(-\sqrt{\beta^2 - \omega_0^2} t) \right] \text{ as we have already calculated before.}$$

ii) Particular solution: Since RHS is a trigonometric function with the form $F_0 \cos \omega t$ we try

$$x_p(t) = D \cos(\omega t - \delta), \quad \dot{x}_p(t) = -D\omega \sin(\omega t - \delta), \quad \ddot{x}_p(t) = -D\omega^2 \cos(\omega t - \delta)$$

If we plug these solutions in the original differential equation:

$$-D\omega^2 \cos(\omega t - \delta) - 2\beta D\omega \sin(\omega t - \delta) + \omega_0^2 D \cos(\omega t - \delta) = F_0 \cos \omega t$$

$$\cos \omega t \cos \delta + \sin \omega t \sin \delta \quad \sin \omega t \cos \delta - \cos \omega t \sin \delta \quad \cos \omega t \cos \delta + \sin \omega t \sin \delta$$

$$-D\omega^2 \cos \omega t \cos \delta - D\omega^2 \sin \omega t \sin \delta - 2\beta D\omega \sin \omega t \cos \delta + 2\beta D\omega \cos \omega t \sin \delta + \omega_0^2 D \cos \omega t \cos \delta + \omega_0^2 D \sin \omega t \sin \delta - F_0 \cos \omega t = 0$$

$$\cos \omega t \{ -D\omega^2 \cos \delta + 2\beta D\omega \sin \delta + \omega_0^2 D \cos \delta - F_0 \} + \sin \omega t \{ -D\omega^2 \sin \delta - 2\beta D\omega \cos \delta + \omega_0^2 D \sin \delta \} = 0$$

$$\cos \omega t \{ D \cos \delta [\omega_0^2 - \omega^2] + 2\beta D\omega \sin \delta - F_0 \} + \sin \omega t \{ D \sin \delta [\omega_0^2 - \omega^2] - 2\beta D\omega \cos \delta \} = 0$$

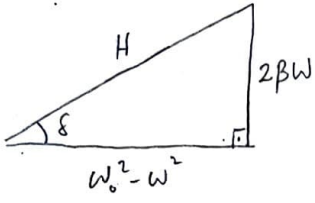
If we write the last equation in a more compact form:

$$\left\{ \begin{aligned} F_0 - D \left[(\omega_0^2 - \omega^2) \cos \delta + 2\beta\omega \sin \delta \right] \cos \omega t \\ - D \left[(\omega_0^2 - \omega^2) \sin \delta - 2\beta\omega \cos \delta \right] \sin \omega t = 0 \end{aligned} \right\} \begin{array}{l} \text{sin and cos functions are linearly} \\ \text{independent. Therefore, coefficient of each} \\ \text{term has to vanish to satisfy this equation} \\ \text{for all } t \text{ values.} \end{array}$$

• From $\sin(\omega t)$ term:

$$D \left[(\omega_0^2 - \omega^2) \sin \delta - 2\beta\omega \cos \delta \right] = 0 \Rightarrow \frac{\sin \delta}{\cos \delta} = \frac{2\beta\omega}{\omega_0^2 - \omega^2} \Rightarrow \tan \delta = \frac{2\beta\omega}{\omega_0^2 - \omega^2}$$

$\Rightarrow$



$$(2\beta\omega)^2 + (\omega_0^2 - \omega^2)^2 = H^2 \rightarrow H = \sqrt{4\beta^2\omega^2 + (\omega_0^2 - \omega^2)^2}$$

$$\Rightarrow \sin \delta = \frac{2\beta\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} \text{ and } \cos \delta = \frac{\omega_0^2 - \omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}$$

• From $\cos(\omega t)$ term:

$$\frac{F_0}{m} - D \left[(\omega_0^2 - \omega^2) \cos \delta + 2\beta\omega \sin \delta \right] = 0$$

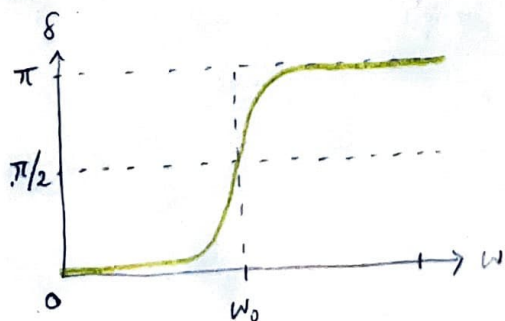
$$\Rightarrow D = \frac{F_0/m}{(\omega_0^2 - \omega^2) \cos \delta + 2\beta\omega \sin \delta} = \frac{F_0/m}{\frac{(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2)}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} + \frac{2\beta\omega \cdot 2\beta\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}}$$

$$= \frac{F_0/m}{\sqrt{[(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2]^2}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}$$

Therefore, the particular solution, finally, is:

$$x_p(t) = \frac{\underbrace{F_0}_{A = F_0/m}}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} \cos(\omega t - \delta) \quad \text{with} \quad \delta = \arctan\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

★ The quantity δ represents the phase difference between the driving force and the resultant motion. It explicitly depends on the angular frequencies of the oscillator (in absence of a driving force) and the driving force (we will turn back to this soon)



The general solution is:

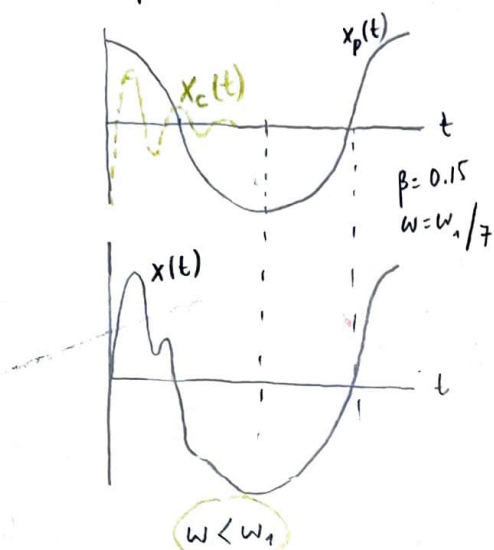
$$x(t) = x_c(t) + x_p(t)$$

- x_c oscillates with ω_0
- x_p oscillates with ω
- δ phase diff. b/w osc. freq of x_c and x_p

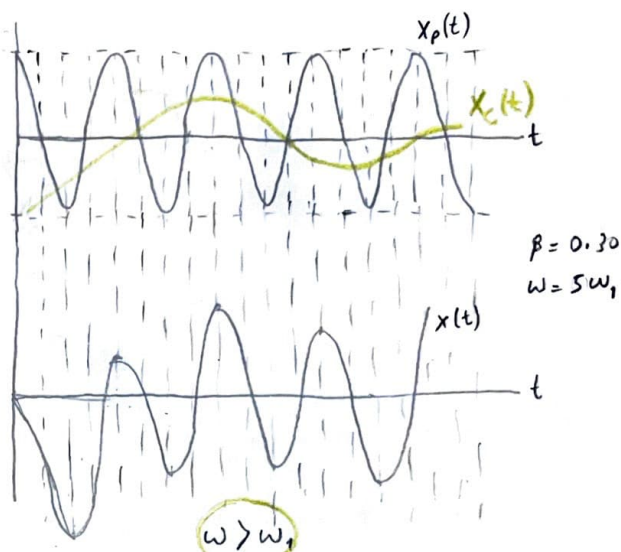
Here $x_c(t)$ represents transient effects; the terms contained in $x_c(t)$ damp out because of the factor $e^{-\beta t}$. The term $x_p(t)$ represents the steady-state effects and contains all the information for t large compared to $1/\rho$:

$$x(t) \gg 1/\rho \approx x_p(t)$$

- ★ The details of the motion during the period before the transient effects have disappeared strongly depend on the oscillator's conditions at that time that the driving force is first applied and also on the relative magnitudes of the driving frequency ω and the damping frequency ω_1 .



Transient response of the oscillator greatly distorts the sinusoidal shape of the forcing function during the time interval immediately after the application of the driving force.



The effect is a modulation of the forcing function with little distortion of the high frequency sinusoidal oscillations.

Resonance Phenomena

Question: which value of the angular frequency of the driving force makes the amplitude of the steady-state effects maximum?

$$\left. \frac{dD}{d\omega} \right|_{\omega=\omega_R} = 0 \quad \text{where } \omega_R \text{ is the angular resonance frequency.}$$

$$D = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} \Rightarrow \frac{dD}{d\omega} = \frac{F_0}{m} \left(-\frac{1}{2}\right) \left[(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2\right]^{-3/2} \left[(-2\omega)2(\omega_0^2 - \omega^2) + 8\beta^2\omega\right]$$

To find ω_R :

$$\frac{F_0}{m} \left(-\frac{1}{2}\right) \left\{ \frac{1}{(\omega_0^2 - \omega_R^2)^2 + 4\beta^2\omega_R^2} \right\}^3 \left[(-4\omega_R)(\omega_0^2 - \omega_R^2) + 8\beta^2\omega_R \right] = 0$$

$$8\beta^2\omega_R = 4\omega_R(\omega_0^2 - \omega_R^2)$$

$$2\beta^2 = \omega_0^2 - \omega_R^2$$

$$\omega_R = \sqrt{\omega_0^2 - 2\beta^2}$$

First compute ω_0 , ω_1 and ω_R go to (*)

What does ω_R tell us?

- i) ω_R is lowered as the damping coefficient is increased.
- ii) No resonance occurs if $\beta > \omega_0/\sqrt{2}$ because ω_R becomes imaginary and it has no physical meaning in this context. Therefore, D monotonically decreases as we increase ω .

Comparison of oscillation frequencies

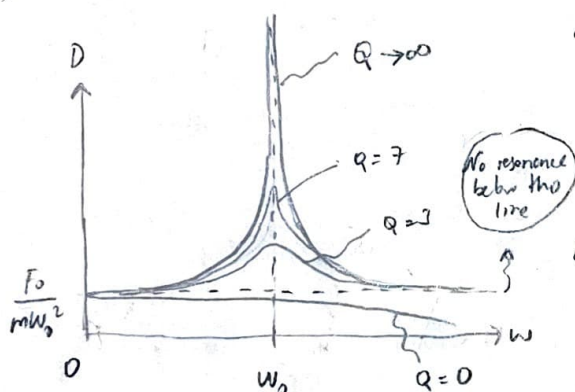
(*)

- Free oscillations without damping: $\omega_0 = \sqrt{k/m}$
- Free oscillations with damping: $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$ with $\beta = b/2m$
- Driven oscillations with damping: $\omega_R = \sqrt{\omega_0^2 - 2\beta^2}$

→ Note that $\omega_0 > \omega_1 > \omega_R$ for a given oscillating system with a fixed k and m .

iii) If there is no damping: $\beta = 0 \Rightarrow \omega_R = \omega_0$ and $D \rightarrow \infty$ at $\omega = \omega_0$

iv) Let us sketch D w.r.t. ω : To do this let's consider following facts:



a) when $\omega = 0$, $D = F_0 / m \omega_0^2 = F_0 / k$

★ It means that the driving force is next oscillatory.
 $\Rightarrow$ There is a constant force F_0 applying on the block.
 It is intuitive since the system gets into equilibrium.
 $F_0 = kx \rightarrow x = F_0/k$

b) If $\beta = 0 \Rightarrow$ No damping $\Rightarrow \omega_R = \omega_0$

$$D = \frac{F_0}{m} \frac{1}{|\omega_0^2 - \omega^2|}$$

c) If $\beta > \omega_0/\sqrt{2}$: First consider $\beta = \omega_0/\sqrt{2}$

$$D = \frac{F_0}{m} \frac{1}{\omega_0^4 - 2\omega_0^2\omega^2 + \omega^4 + \frac{\omega_0^2}{2}\omega^2} = \frac{F_0}{m} \frac{1}{\omega_0^4 + \omega^4}$$

D monotonically decreases: No resonance occurs!
 and this will happen for any other value of $\beta > \omega_0/\sqrt{2}$ since the expression inside the square root is completely positive. However, when $\beta < \omega_0/\sqrt{2}$, let's say $\beta = \omega_0/2$,

$$D = \frac{F_0}{m} \frac{1}{\omega_0^4 - 2\omega_0^2\omega^2 + \omega^4 + \omega_0^2\omega^2} = \frac{F_0}{m} \frac{1}{\omega_0^4 + \omega^4 - \omega_0^2\omega^2}$$

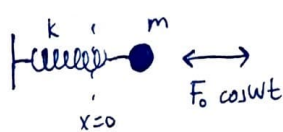
$$= \frac{F_0}{m} \frac{1}{\omega_0^4 + \omega^2(\omega^2 - \omega_0^2)}$$

when $\omega^2 < \omega_0^2$
 the contribution to the denominator will be negative
 $\Rightarrow D \uparrow$

It is conventional to describe the degree of damping in an oscillating system in terms of a "quality factor" $Q \equiv \omega_R / 2\beta$.

unitless

DRIVEN OSCILLATION (STEADY-STATE SOLUTION)



$$\frac{b}{2m} \equiv \beta \quad \omega_0^2 = \frac{k}{m}$$

Taking the derivatives of $x_p = D \cos(\omega t - \delta)$ requires a lot of effort. Instead let us switch to complex plane and rewrite (*) as

$$\ddot{z} + 2\beta \dot{z} + \omega_0^2 z = \frac{F_0}{m} e^{i\omega t}$$

and instead of $x_p(t) = D \cos(\omega t - \delta)$ let us consider

$$z(t) = A e^{i(\omega t - \delta)} \quad \dot{z} = iA\omega e^{i(\omega t - \delta)} \quad \ddot{z} = -A\omega^2 e^{i(\omega t - \delta)}$$

• We want to determine A and δ

• If we plug $z, \dot{z}$ and $\ddot{z}$ into D.E.

$$(-\omega^2 + 2\beta i\omega + \omega_0^2) A e^{i(\omega t - \delta)} = \frac{F_0}{m} e^{i\omega t}$$

$$(-\omega^2 + 2\beta i\omega + \omega_0^2) A = \frac{F_0}{m} e^{i\delta} = \frac{F_0}{m} (\cos \delta + i \sin \delta)$$

$$(-\omega^2 + \omega_0^2) A = \frac{F_0}{m} \cos \delta \quad \text{and} \quad 2\beta \omega A = \frac{F_0}{m} \sin \delta \quad (1)$$

$$\frac{(1)}{(2)} = \tan \delta = \frac{2\beta \omega}{\omega_0^2 - \omega^2} \quad \text{phase}$$

$$(1)^2 + (2)^2 = \frac{F_0^2}{m^2} = A^2 \left[(\omega_0^2 - \omega^2)^2 + (2\beta \omega)^2 \right]$$

$$\Rightarrow A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\beta \omega)^2}} \quad \text{amplitude}$$

• Back to the real world:

$$\text{Re}[z(t)] = A \cos(\omega t - \delta) \quad \text{or } x_p(t) : \text{particular solution or "steady-state" solution.}$$

Some observations for A:

- 1) F_0 is in the numerator; $F_0 \uparrow, A \uparrow$
- 2) β is in the denominator; $\beta \uparrow, A \downarrow$
- 3) $\omega \ll 1$: $A = \frac{F_0}{m\omega_0^2} = \frac{F_0}{k}$

! start considering this case with pure intuition. what would A and δ be?
 $F = kx$: equilibrium.
 If $F(t)$ is so slow kx and $F(t)$ must cancel each other all the time.
 $\delta \rightarrow 0$

We have seen that the equation of motion then can be written:

$$m\ddot{x} = -kx - b\dot{x} + F_0 \cos \omega t$$

$$\ddot{x} + 2\beta \dot{x} + kx = F_0 \cos \omega t \quad (*)$$

and we know that the general solution can be written as

$$x(t) = x_c(t) + x_p(t)$$

$x_c(t)$: We considered x_c in damped osc. section

$x_p(t)$: We guessed $x_p(t) = D \cos(\omega t + \delta)$ and plugged it into the diff. eq. to obtain D and δ .

★ It is very important to remember and observe that the complementary solution had to "adjustable" x_0, v_0 , i.e., it depends on the initial conditions.

However, here we see that $x_p(t)$ has nothing to do with the initial conditions.

The reason is that this is a steady-state solution which means the system does not even "remember" anymore what the situation was at $t=0$, it had lost all its memory.

4) What if $\omega = \omega_0$? (resonance frequency)

$$A = \frac{F_0}{m\omega_0^2(2\beta)} = \frac{F_0 \omega_0}{m\omega_0^2 2\beta} = \frac{F_0}{k} Q$$

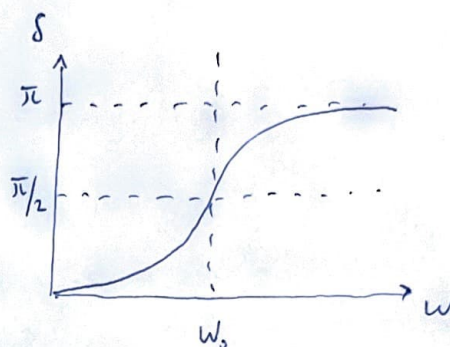
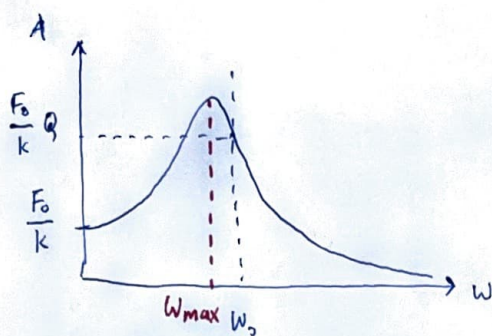
Note that F_0/k was the amplitude in very long frequency and when we drive it in the resonance freq. it is Q times higher.

$$\delta = \pi/2$$

5) $\omega \rightarrow \infty$: Everything goes so fast that the object has no time to follow the driver.

$\Rightarrow A \rightarrow 0$: no amplitude

$\delta \rightarrow \pi$



$$\left. \frac{dA}{d\omega} \right|_{\omega=\omega_{\max}} = 0 \rightarrow \omega_{\max} = \sqrt{\omega_0^2 - 2\beta^2} = \omega_0 \sqrt{1 - \frac{2\beta^2}{\omega_0^2}} = \omega_0 \sqrt{1 - \frac{(2\beta)^2}{2\omega_0^2}} = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$

$$\boxed{\omega_{\max} = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}} \quad A_{\max}$$

The frequency for the kinetic energy resonance:

What we have discussed recently was the frequency for the amplitude resonance. Now we will calculate the value ω which T is maximum.

$$T = \frac{1}{2} m \dot{x}^2$$

$$x_p(t) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} \cos(\omega t - \delta)$$

$$\dot{x} = \frac{-F_0\omega/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} \sin(\omega t - \delta)$$

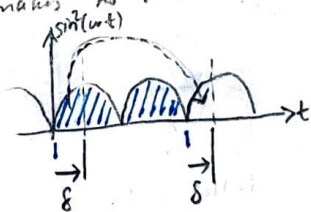
$$T = \frac{1}{2} m \frac{F_0^2 \omega^2}{m^2} \frac{\sin^2(\omega t - \delta)}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2} = \frac{F_0^2 \omega^2}{2m} \frac{\sin^2(\omega t - \delta)}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

To obtain a value of T independent of time (or in the other words, we are especially interested in how much energy does the system have over one complete oscillation:

$$\langle T \rangle = \frac{F_0^2 \omega^2}{2m} \frac{1}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2} \langle \sin^2(\omega t - \delta) \rangle$$

$$\langle \sin^2(\omega t - \delta) \rangle = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \sin^2(\omega t - \delta) dt = \frac{1}{2}$$

→ If this integral is taken over a period of $\sin^2$ function the phase δ makes no sense:



$$\Rightarrow \langle \sin^2(\omega t - \delta) \rangle = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \sin^2(\omega t) dt = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \frac{1 - \cos(2\omega t)}{2} dt = \frac{\omega}{2\pi \cdot 2} \left[t - \frac{\sin(2\omega t)}{2\omega} \right]_0^{2\pi/\omega}$$

$$= \frac{\omega}{2\pi \cdot 2} \left[\frac{2\pi}{\omega} - \frac{\sin(2\omega \cdot \frac{2\pi}{\omega})}{2\omega} - \left(0 - \frac{\sin(0)}{2\omega} \right) \right] = \frac{1}{2}$$

• Therefore,

$$\langle T \rangle = \frac{F_0^2 \omega^2}{4m} \frac{1}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

• The value ω which makes $\langle T \rangle$ maximum

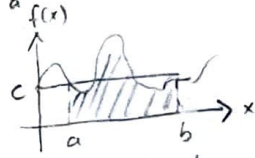
$$\frac{d[\dots]}{d\omega} = -\frac{2\omega^2(\omega_0^2 - \omega^2)}{4m[(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2]^2} + \frac{4\beta^2\omega}{4m[(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2]^2}$$

$$\omega_0^4 - 2\omega_0^2\omega^2 + \omega^4 + 4\beta^2\omega^2 = -2\omega^2\omega_0^2 + 2\omega^4 + 4\beta^2\omega^2$$

$\omega^4 = \omega_0^4 \rightarrow \boxed{\omega = \omega_0}$! Kinetic energy resonance occurs at the natural frequency of the system.

Mean value theorem for integral:

$$\frac{1}{b-a} \int_a^b f(x) dx = c$$



$$(b-a)c = \int_a^b f(x) dx$$

★ Amplitude resonance occurs at $\omega_R = \sqrt{\omega_0^2 - 2\beta^2}$ whereas the kinetic energy resonance occurs at ω_0 . Since the potential energy is proportional to the square of amplitude, the potential energy resonance must also occur at $\sqrt{\omega_0^2 - 2\beta^2}$. That T and U resonate at different frequencies is a result of the fact that the damped oscillator is not a conserved system. Energy is continually exchanged with the driving mechanism, and energy is being transferred to the damping medium.

CENTRAL FORCE MOTION

The motion of a system consisted two bodies affected by a force directed along the line connecting the centers of these bodies.

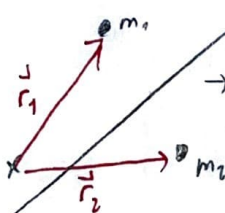
It is an important problem.

→ It is completely solvable.

→ Explains the motion of celestial bodies.

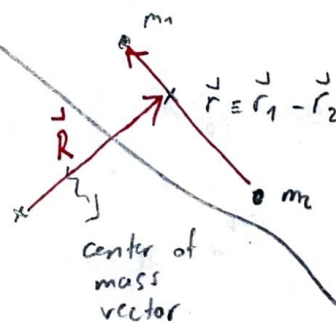
→ Explains certain 2 body nuclear interactions (e.g. scattering of α particles from a nuclei)

Reduced mass:



→ we need 6 real numbers to describe the positions of the objects

Alternatively:



→ Again we end up with 6 real quantities

29.9.22

Exercises:

- ① Find the velocity of the block after it moves from rest a distance x_0 down the inclined plane with $\theta = 30^\circ$.

What if θ changes in time with a rate

$$\ddot{x} = g \sin \theta \quad \dot{x}(t) = g t \sin \theta + C = g t \sin \theta + V_0 \quad x(t) = \frac{1}{2} g t^2 \sin \theta + V_0 t + x_0$$

multiply this with $2\dot{x}$

$$2\dot{x}\ddot{x} = 2\dot{x}g \sin \theta \rightarrow \frac{d}{dt}(\dot{x}^2) = 2\frac{dx}{dt}g \sin \theta \rightarrow \int_0^{V_0^2} d(\dot{x}^2) = 2g \sin \theta \int_0^{x_0} dx \rightarrow V_0^2 = 2g \sin \theta x_0 \rightarrow V_0 = \sqrt{2g \sin \theta x_0}$$

- ② If the coeff. of static friction between the block and the plane is μ_s at what angle θ will the block start sliding if it is initially at rest?

$$f_s \sim N \rightarrow f_s = \mu_s N$$

$$\bullet \text{ y-direction: } F_g \cos \theta + N = 0$$

$$\bullet \text{ x-direction: } -f_s + F_g \sin \theta = m\ddot{x}$$

f_s will keep cancelling out $-f_s + F_g \sin \theta$ and keeping the block at rest as we increase the angle θ . However, after some critical angle θ' the f_s will not be able to keep the block at rest. At that angle θ' , f_s becomes:

$$f_s(\theta = \theta') = f_{\max} = \mu_s N = \mu_s F_g \cos \theta'$$

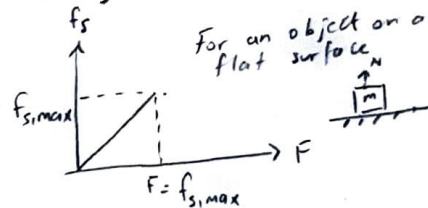
$$\rightarrow m\ddot{x} = F_g \sin \theta' - f_{\max} = F_g \sin \theta' - \mu_s F_g \cos \theta'$$

$$\Rightarrow m\ddot{x} = m g \sin \theta' - \mu_s m g \cos \theta' \rightarrow \ddot{x} = g(\sin \theta' - \mu_s \cos \theta')$$

- Just before block starts to slide, the acceleration $\ddot{x} = 0$.

$$\sin \theta' - \mu_s \cos \theta' = 0 \rightarrow \sin \theta' = \mu_s \cos \theta' \rightarrow \tan \theta' = \mu_s \rightarrow \theta' = \arctan(\mu_s)$$

- ③ After the block



$$\sin \theta = \frac{h}{\ell}$$

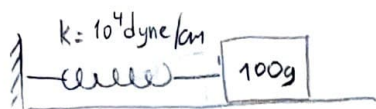
$$mgh = \frac{1}{2} m v^2$$

$$2xg \sin \theta =$$

Exercises:

27.10.22

①



The mass is displaced 3cm and released from rest.

a) calculate natural frequency ν_0 and period.

b) The total energy

c) maximum speed.

Give your results in SI units.

1 dyne = 10^{-5} N

$$a) \omega_0 = \sqrt{\frac{k}{m}} \quad \omega_0 = 2\pi\nu_0 \rightarrow \nu_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{10^4 \frac{\text{dyne}}{\text{cm}} \cdot \frac{10^{-5} \text{ N}}{1 \text{ dyne}} \cdot \frac{100 \text{ g}}{1 \text{ m}}}{100 \text{ g} \cdot \frac{1 \text{ kg}}{1000 \text{ g}}}} = \frac{1}{2\pi} \sqrt{\frac{10 \frac{\text{N}}{\text{m}}}{\frac{1}{10} \text{ kg}}}$$

$$= \frac{1}{2\pi} \sqrt{100 \frac{\text{kg m}}{\text{s}^2 \text{ kg}}} = \frac{10}{2\pi} \frac{1}{\text{s}} = \frac{5}{\pi} \text{ Hz}$$

$$T_0 = \frac{1}{\nu_0} = \frac{\pi}{5} \text{ s}$$

$$b) E_T = \frac{1}{2} k X_{\text{max}}^2 = \frac{1}{2} \cdot 10^4 \frac{\text{dyne}}{\text{cm}} \cdot \left(3 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \right)^2 \cdot \frac{10 \text{ dyne}}{1 \text{ N}} = \frac{1}{2} \frac{\text{N}}{\text{m}} (0.03 \text{ m})^2 \cdot 10 = \frac{1}{2} 3^2 \frac{\text{N m}^2}{\text{m}} = \frac{9}{2} \text{ J} = 4.5 \text{ J}$$

$$c) \frac{1}{2} m V_{\text{max}}^2 = E_T \Rightarrow \sqrt{\frac{2 E_T}{m}} = V_{\text{max}} \Rightarrow V_{\text{max}} = \sqrt{\frac{2 \times 4.5 \text{ J}}{0.1 \text{ kg}}} = \sqrt{\frac{9 \text{ m}^2}{0.1 \text{ s}^2}} = \sqrt{90} \frac{\text{m}}{\text{s}} = 3\sqrt{10} \frac{\text{m}}{\text{s}}$$

②

Assume that the motion in the problem above takes place in a resisting medium. After oscillating for 10s, the max amplitude decreases to half the initial value. Calculate,

a) The damping parameter β

b) The frequency (compare with the undamped frequency) ($\ln 2 \approx 0.7$)

$$a) A_0 e^{-\beta t} = \frac{1}{2} A_0 \rightarrow \ln[e^{-\beta t}] = \ln\left[\frac{1}{2}\right] \Rightarrow -\beta t = -\ln 2 \rightarrow \beta = \frac{\ln 2}{t}$$

$$\Rightarrow \beta = \frac{\ln 2}{10 \text{ s}} \rightarrow \beta = \frac{\ln 2}{10} \text{ s}^{-1}$$

$$b) \omega_1 = \sqrt{\omega_0^2 - \beta^2}$$

$$\omega_0 = 2\pi\nu_0 = 2\pi \frac{5}{\pi} \text{ Hz} = 10 \text{ s}^{-1}$$

$$\Rightarrow \omega_1 = \sqrt{100 \frac{1}{\text{s}^2} - \frac{(\ln 2)^2}{100} \frac{1}{\text{s}^2}} = \frac{10}{\text{s}} \sqrt{1 - \frac{0.49}{10000}} = \frac{10}{\text{s}} \sqrt{1 - 0.49 \times 10^{-4}}$$

$$\nu_1 = \frac{\omega_1}{2\pi} = \frac{5}{\pi} \sqrt{1 - 0.49 \times 10^{-4}} \text{ s}^{-1} \text{ almost the same!}$$

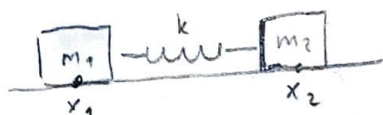
③ The oscillator in Ex 1. is set into motion by giving it an initial velocity of $\frac{1 \text{ cm}}{\text{s}}$ at its equilibrium position. calculate,

- the max displacement
- the max potential energy.

$$\begin{aligned}
 \text{a) } \frac{1}{2} m v^2 &= \frac{1}{2} k x_{\max}^2 \Rightarrow x_{\max} = \sqrt{\frac{m}{k}} v = \sqrt{\frac{m}{k}} v = \sqrt{\frac{100 \text{ g}}{10 \text{ N/m}}} \cdot \frac{1 \text{ cm}}{\text{s}} \\
 &= \sqrt{\frac{1 \text{ kg}}{10}} \cdot \frac{1}{100} \frac{\text{m}}{\text{s}} = \sqrt{\frac{1 \text{ kg}}{10} \cdot \frac{1 \text{ m}}{100 \frac{\text{kg m}}{\text{s}^2}}} \cdot \frac{1}{100} \frac{\text{m}}{\text{s}} \\
 &= \frac{1}{10} \frac{\text{m}}{\text{s}} \cdot \frac{1}{100} = 0.001 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } U_{\max} &= \frac{1}{2} k x_{\max}^2 = \frac{1}{2} \cdot 10 \frac{\text{N}}{\text{m}} \cdot (0.001)^2 \text{ m}^2 \cdot \frac{100 \text{ mN}}{1 \text{ m}} = \frac{1}{2} \cdot 10^{-1} (0.001)^2 \times 10^8 \text{ mN} \\
 &= 2 \times 10^{-5} \text{ J}
 \end{aligned}$$

④



Two masses $m_1 = 100 \text{ g}$ $m_2 = 200 \text{ g}$ slide freely in a horizontal frictionless track and are connected by a spring whose force constant is $k = 0.5 \text{ N/m}$. Find the frequency of the oscillatory motion for this system.

$$\begin{aligned}
 m_1 \ddot{x}_1 &= -k(x_1 - x_2 + \ell) \\
 m_2 \ddot{x}_2 &= -k(x_2 - x_1 - \ell) \rightarrow x_1 = \frac{1}{k} (m_2 \ddot{x}_2 + kx_2 - k\ell)
 \end{aligned}$$

$$\frac{d^2}{dt^2} [m_1 m_2 \ddot{x}_2 + (m_1 + m_2) k x_2] = 0$$

$$\ddot{x}_2 = -\frac{m_1 + m_2}{m_1 m_2} k x_2 \Rightarrow \omega = \sqrt{\frac{m_1 + m_2}{m_1 m_2} k} \quad \text{we obtain the same result for } x_1.$$

If we define a "reduced mass" $\mu = \frac{1}{m_1} + \frac{1}{m_2} = \frac{m_1 + m_2}{m_1 m_2}$

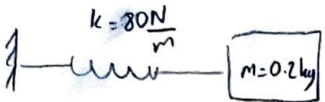
$$\text{We can rewrite } \omega \text{ as } \omega = \sqrt{\frac{k}{\mu}} \Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

This means the system oscillates in the same way as a system consisting of a single mass μ .

- ⑤ The amplitude of lightly damped oscillator decreases by 3.0% each cycle. What percentage of the mechanical energy is lost in each cycle?

$$E = \frac{1}{2} k A^2$$

$$\frac{A_2}{A_1} = \frac{95}{100} = 0.95 \quad \frac{E_2}{E_1} = \frac{A_2^2}{A_1^2} = (0.95)^2 = 0.9025 \Rightarrow 9.75\%$$

- ⑥  The body is subjected to a resistive force given by $-bV$
 $b = 4 \frac{\text{N}}{\text{m s}}$

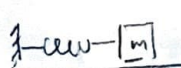
- a) Set up diff eq. for free osc. and find the period

$$\ddot{x} + \frac{b}{m} \dot{x} + \frac{k}{m} x = 0$$

- b) The object is subjected $f(t) = F_0 \sin(\omega t)$ $F_0 = 2\text{N}$ $\omega = 30 \frac{\text{rad}}{\text{s}}$. In steady state what is the amp. of forced os.

$$A(\omega) = \frac{F_0/m}{(\omega_0^2 - \omega^2)^2}$$

- ⑦ Write the equation of motion for a mass m attached to a spring with spring constant k hanging from the ceiling. Show that the equation of motion is the same as

 $\rightarrow$  : would the frequency of the oscillation change?